

Mitchell Zhou

Cybersecurity Engineer at MITRE passionate about Offensive security—to me it's computer magic: understanding systems so deeply you can make them do what they were never intended to. I enjoy doing CVE research, CTFs, and hands-on security projects that bridge theory with real-world impact.

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Education

Carnegie Mellon University

M.S Information Security

08/2024 - 05/2026

GPA: **3.83/4.0**

New York University

B.S Computer Science

09/2020 - 05/2024

GPA: **3.90/4.0**

Work Experience

Cyber New Professional (CNP)

MITRE

06/2026 - Present

40 hr/week

Teaching Assistant - Secure Coding

Carnegie Mellon University

08/2025 - 12/2025

15 hr/week

Achievements

- Led weekly recitation sections of 30+ students, clarifying security concepts such as buffer overflows, memory exploitation, pointer vulnerabilities, and truncation errors
- Created supplemental learning materials, such as homework solution sheets and exam review guides, that will be adopted by future TAs
- Collaborated with professors to design 2 new CTF challenges

Cybersecurity Intern - SOC Analyst

Capital One

06/2025 - 08/2025

40 hr/week

Achievements

- Discovered and reported a publicly exposed test environment via the Bug Bounty program—a **company record for fastest bug bounty find (4 weeks)**—preventing a potential breach and saving ~\$10,000.
- Developed automation pipelines for Indicator of Compromises (IoC) analysis; including mass IoC Scanning, IoC Pivoting, and IoC Scrapping
- Identified and blocked 50+ Indicators of Compromise (IoCs) tied to the North Korean threat group Famous Chollima
- Developed email bombing behavioral detection targeting the BlackSuit ransomware group's TTPs

Cybersecurity Intern

Textron Systems

05/2023 - 8/2023

40 hr/week

Achievements

- Conducted security assessments using SCAP, Nessus, and Splunk for vulnerability scans and monitoring, identifying and mitigating 14+ critical vulnerabilities across 10+ systems
- Developed 20+ PowerShell and Python scripts to automate tasks and enhance operational efficiency

Awards/Projects

MITRE eCTF - 1st Place

01/2026 - 05/2026

Performed voltage-glitching attacks, buffer overflow exploits to extract flash memory dumps, and hardware signal analysis on embedded systems.

CVE Exploit Development

08/2024 - Present

Developed and released 4 PoC exploits for high-severity, CVEs across Git, DOMPurify, WordPress, and Python packages.

CMU Badge Scanner Security

07/2022 - 09/2022

Penetration tested CMU's badge scanner infrastructure, leveraging a Proxmox3 to crack card encryption keys and achieved full card cloning and spoofing.

NYU Email Spoofer

07/2022 - 09/2022

Successfully penetrated NYU email servers and sent verified emails with a spoofed "@nyu.edu" domain.

Organizations

Plaid Parliament of Pwning (PPP)

08/2024 - 05/2026

Member

- CTF competitor on CMU's world-renowned PPP team
- 1st place finish at the MITRE Embedded Capture the Flag (eCTF) competition.
- Developed CTFs on PicoCTF which are used by CMU classes, such as Secure Coding & Ethical Hacking

NYU Osiris Cyber Security Club

05/2022 - 05/2024

CTF Dev Lead

- Competed in 15+ CTF competitions with the NYU SEC team
- Developed web exploitation challenge and reverse engineering challenges to NYU CSAW Hackathon

Certifications/Clearances

DoD Top Secret Clearance (*interim*)

CompTIA Security+

Verification Code: 0ZJ1V73RH1FEQ4K8

Publications

From Blocking to Breaking: Evaluating the Impact of Adblockers on Web Usability

09/2023 - 04/2025

Priority Based Routing and Link Scheduling for Cognitive Radio Networks

04/2019 - 06/2019